Municipality Library System

Deliverable #1

Group – 3

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1. Introduction

1.1. Project Overview

The current municipal library system relies on manual catalog cards, which, while simple and easy to use, lacks accessibility, accuracy, and efficiency. Users face difficulties in tracking and monitoring books over time, and staff are prone to human errors during record-keeping.

The proposed solution is a digital library management system that provides real-time tracking of all loaned books and reservations, minimizes manual processing errors, and supports multi-user access. The system supports remote access, scalability and reporting

1.2. Recommendation

After analyzing the current system and evaluating opportunities for improvement, it is recommended to implement a digital library management system. This system will:

- Provide precise tracking of all loaned books and inventory.
- Automate processes to reduce manual filing and data entry mistakes.
- Enable scalability to accommodate future growth in the library's catalog.
- Offer remote access to members and allow simultaneous multi-user operations.

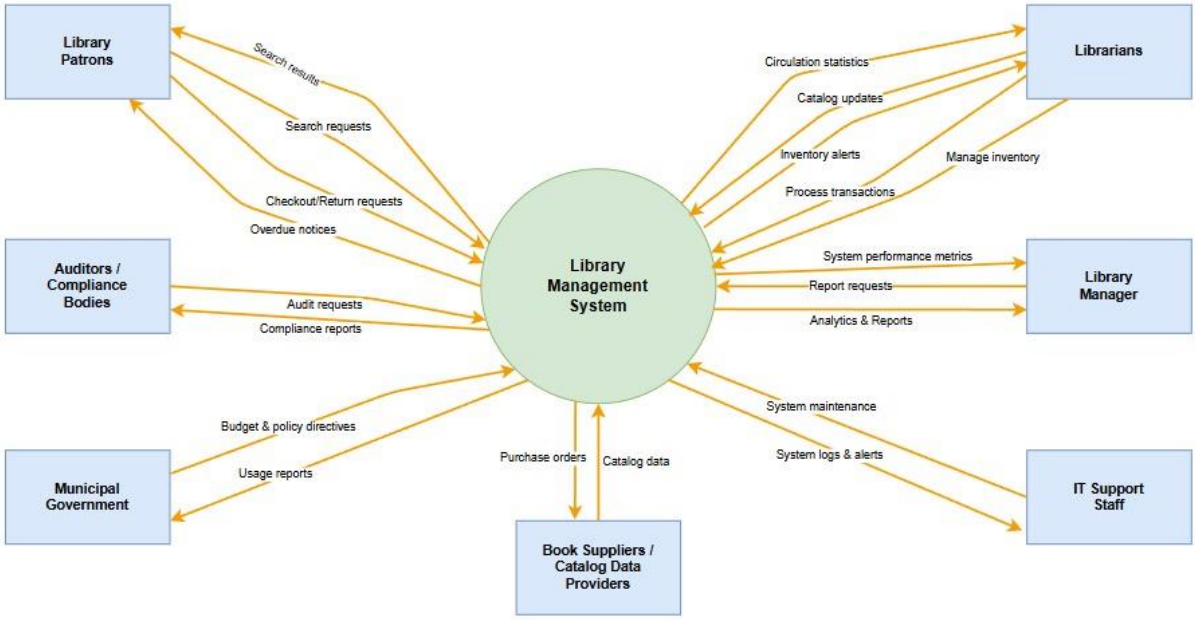
Adopting this system will enhance operational efficiency, reduce errors, and provide data-driven insights for informed decision-making, ensuring the library can meet the needs of both staff and patrons effectively.

2. System Description

2.1 Current System Environment

2.1.1 Description of Current System:

The library currently operates using a manual card catalog system with paper-based member records and handwritten checkout cards. The system includes physical index cards organized alphabetically by title, author, and subject, stored in catalog drawers. All transactions are processed manually during library operating hours, requiring in-person visits by patrons.



2.1.2 Analysis of Current System

Strengths:

- Simple and intuitive - no technical training required
- Technology independent - operates without electricity or internet
- Low initial cost and easy physical expansion

Weaknesses:

- Limited accessibility - no remote access, difficult for users with disabilities
- Time-consuming manual searches and labor-intensive processing
- Error-prone record keeping with no centralized tracking
- Limited reporting capabilities and single-user bottleneck

Opportunities:

- Enhanced tracking visibility - real-time status of all loans and reservations
- Minimize manual errors through automated workflows
- 24/7 remote catalog access and multi-user accessibility
- Data-driven insights for collection management and resource allocation

Threats:

- Staff resistance to change and technical skill gaps
- Municipal budget constraints limiting scope
- System downtime risks and data migration challenges

2.2 Proposed System Scope & Objectives

2.2.1 System Scope

In Scope:

- **Digital Book Catalog** - searchable database with real-time availability
- **User Management** - member registration and account management
- **Loan Tracking** - automated checkout/return with transaction history
- **Reservation System** - online holds with waiting list management
- **Reporting Tools** - circulation statistics, usage analytics, exportable reports
- **Automated Notifications** - email/SMS for due dates and overdue items
- **Remote Access** - 24/7 web-based interface for patrons
- **Multi-User Support** - concurrent access for staff and patrons
- **Inventory Management** - catalog editing and collection assessment tools
- **Data Migration** - conversion of paper records to digital format
- **Staff Training** - comprehensive program for all library staff

Out of Scope:

- Building modifications or renovations
- Service expansions (e-books, inter-library loans, event management)
- External integrations (regional networks, payment processing)
- Advanced features (AI recommendations, mobile apps, RFID, self-checkout kiosks)

2.2.2 System Business Objectives

Objectives:

- Improve operational efficiency and reduce manual work
- Provide 24/7 remote access to catalog
- Reduce errors through automation
- Generate reports and analytics for better decisions
- Improve patron satisfaction with faster service
- Support future collection growth

Constraints:

- Limited municipal budget
- Staff need training due to varying technical skills
- Library must stay open during implementation
- Must complete within current fiscal year
- Must use existing IT infrastructure

- Must protect patron privacy and data

3. Feasibility Assessments.

3.1. Economic Feasibility.

The economic feasibility of the proposed digital library management system evaluates whether the expected benefits justify the cost of implementation. While a formal numerical cost–benefit analysis is not required at this stage, a qualitative assessment demonstrates strong economic justification through both tangible and intangible benefits.

The project requires initial investment in system development or acquisition, staff training, data digitization, and ongoing maintenance. These costs must be weighed against the operational improvements the system provides. From a tangible benefit perspective, the system directly reduces inefficiencies present in the current manual processes. Automating book records, loan tracking, and member information significantly reduces human error, which lowers the time and resources spent correcting mistakes. Increased productivity and faster processing of loans, returns, and reservations enable staff to serve more members within the same working hours.

The removal of physical card catalogs reduces expenses associated with printing, storage, and manual record maintenance. Additionally, the reduction in staff time required per transaction allows the library to maintain service levels without increasing labor costs, creating measurable operational savings.

The system also provides important intangible benefits that strengthen its economic value. Digital access to accurate and timely information improves decision-making for staff and convenience for library members. Reliable book availability and reservation tracking increase public trust in the system. A modern digital infrastructure enhances the library's public image as an efficient and forward-looking institution. Improved user satisfaction encourages continued community engagement and long-term use of library services, indirectly supporting the library's sustainability and relevance.

Taken together, these tangible cost savings and intangible service improvements provide a strong qualitative economic justification for the project. Even without formal financial modeling, the balance of benefits clearly supports investment in the digital system, particularly when implemented in a phased and budget-conscious manner.

3.2. Technical Feasibility.

The proposed digital library management system is technically feasible because it relies on mature, widely available technologies such as database systems, web interfaces, and secure digital storage. The system can operate on standard hardware and does not require specialized infrastructure. Both local and cloud-based deployment options are viable, and existing backup and security practices can be applied to protect library data. The system can also be designed to scale in the future without major redesign.

However, several technical difficulties must be considered. Migrating data from the card catalog to a digital format presents risks of data errors or loss if not carefully validated. During the transition period, integrating manual and digital workflows may temporarily increase operational complexity. Staff adaptation to the new system may require training and user-friendly design. Additionally, maintaining data security, privacy, and system reliability is essential to prevent service interruptions.

Despite these challenges, the required technology is accessible and well understood. With proper planning, testing, and training, the technical risks are manageable, making the project technically achievable.

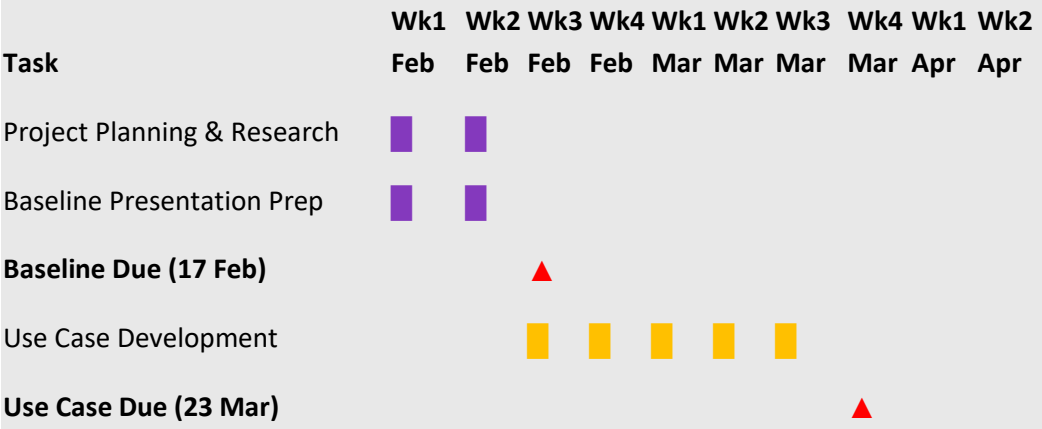
3.3. Operational Feasibility.

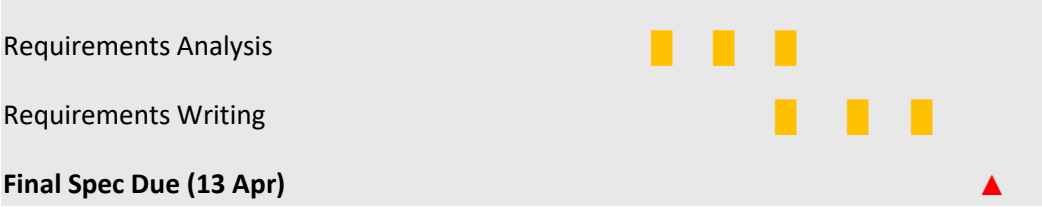
3.4. Legal and Contractual Analysis.

3.5. Political Analysis.

3.6. Schedule.

3.6.1. High Level Project Schedule (Gannt Overview).





4. Management Issues

4.1. Team Configuration

Aakash Yogabalu (Project Lead & System Architect): Developed the Context Diagram and Benefits analysis. Responsibilities include overseeing the development lifecycle, managing the master schedule, and ensuring technical architecture aligns with project goals.

Md Imtiaz Ahmed (Business Analyst): Defined Business Objectives and strategic goals. Responsibilities include justifying the project's scope and acting as the bridge between client business needs and the technical team.

Abdul Gafoor (Lead System Analyst): Conducted the Current System Analysis (SWOT). Responsibilities include documenting legacy workflows and performing gap analysis to address manual process weaknesses.

Asamu Kabir (Scope & Implementation Manager): Defined Stakeholders, Scope, and Constraints. Responsibilities include managing project boundaries to prevent scope creeping and ensuring adherence to budget and operational constraints.

4.2. Communication Plan Internal Collaboration:

Weekly Stand-ups: Hold the end of the week to track progress and assign tasks.

Daily Coordination: Instant messaging (WhatsApp/Teams) is used for quick queries.

Documentation: A shared Google Drive hosts all project files for real-time collaborative editing.

External Reporting:

The Project Lead is responsible for submitting deliverables and status reports to the project sponsor at agreed-upon intervals.

4.3. Other Project-Specific Topics Operational Continuity:

To maintain services, a "rolling migration" strategy will be used, digitizing books as they are returned rather than requiring a full system shutdown.

Skill Gap Management: Due to varying technical skills among staff, specific resources have been allocated for designing a simplified user interface and creating training manuals.